



German-born, Professor Hans Krebs emigrated to England in 1933. He is shown at work in the laboratory at Sheffield University where he was based from 1935 to 1954 and carried out much of his research.

IN JANUARY, ITALIAN physicists **Carlo Perrier** (1866–1948) and **Emilio Segrè** (1905–89) reported on a **new artificial element** formed by atomic reaction. They had made **technetium** in a radioactive-contaminated part of a cyclotron (see 1931).

In February, American chemist **Wallace Carothers** patented his **new chemical polymer**—polyamide 6-6. By 1938, the DuPont Company had called this polymer **nylon** and was

“ A SPECIES IS A STAGE IN PROGRESS, NOT A STATIC UNIT. ”

Theodosius Dobzhansky, in *Genetics and the Origin of Species*, 1937

using it commercially to make toothbrush bristles.

In 1927, French-American engineer **Eugene Houdry** (1892–1937) had invented a way of **cracking petroleum from**

crude oil using a silica-alumina based catalyst. Sun Oil started up the first petroleum cracking unit using the invention in March 1937.

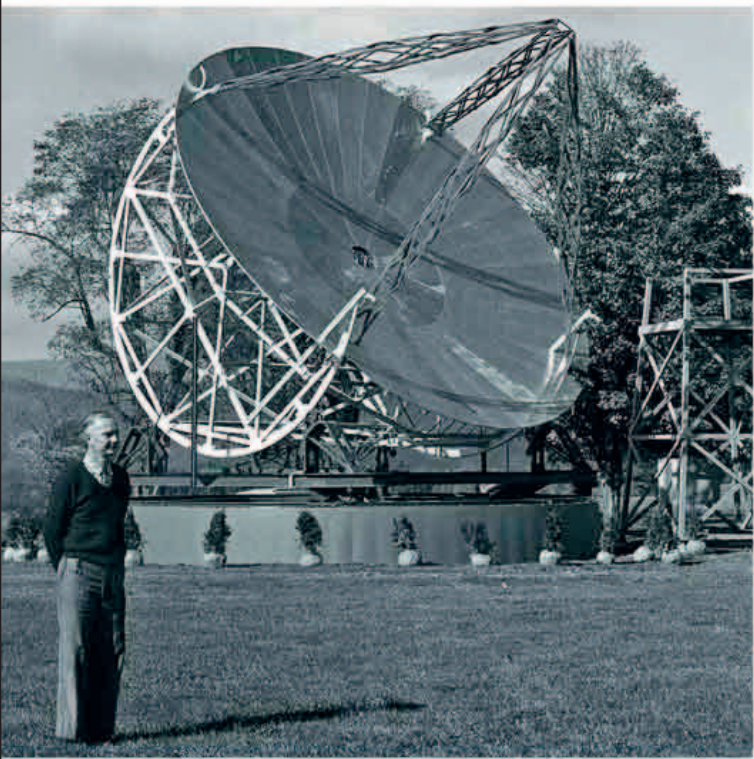
In September, American electrical engineer **Grote Reber** (1911–2002) built the **first radio telescope**.

German-born biochemist **Hans Krebs** found that **citric acid** could keep cells alive and showed that it was a key intermediary in a process that provided cells with energy. The process became known as the **citric acid, or Krebs, cycle**.

Ukrainian biologist **Theodosius Dobzhansky** (1900–75) published *Genetics and the Origin of Species*. In it he showed how Darwin's natural selection (see 1859) could be explained in terms of the genetic makeup of populations and helped lay the foundations of **modern evolutionary biology**.

The first radio telescope

American **Grote Reber** built his radio telescope in his back yard. He used it to confirm that radio signals could be detected from space.



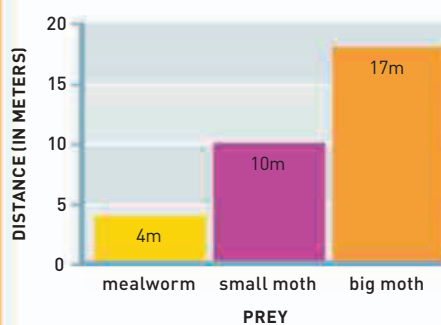
The coelacanth lives in the deep ocean waters and although it was known to local fishermen, paleontologists knew it only as a fossil.

AMERICAN ZOOLOGIST DONALD GRIFFIN (1915–2003) was studying the migratory behavior of bats, but could not understand how they managed to navigate in the dark. In the 18th century, Italian biologist **Lazzaro Spallanzani** (1729–1799) had demonstrated that bats could fly around objects when deprived of sight—but not if their ears were blocked. Working with the American neurologist **Robert Galambos** (1914–2010), Griffin used a special ultrasound microphone to discover that **bats emit high-pitched sounds** that are beyond the range of human hearing. Griffin's theory was that the animals were using a form of sonar, listening for echoes of their sounds as they bounced off obstacles or prey. At first, his theory was derided, but by 1944 the accepted phenomenon was called **echolocation**.

The German chemist **Otto Hahn** (1879–1968) was

experimenting with chemical transmutation—how one element could change into another through a nuclear reaction. Transmutation had been first demonstrated 20 years earlier by New Zealand-born physicist **Ernest Rutherford** (see 1916–17). But scientists believed that there were limits to what could be achieved, and that smashing the heaviest atoms, such as those of uranium, could not make much lighter atoms. However, toward the end of the year, Hahn reported that he had achieved just that—he **split uranium to produce barium**, a process known as **nuclear fission**. The Italian physicist Enrico Fermi had achieved a similar process—although he thought he had synthesized a new element (see 1934). Later, Fermi would oversee the first fission chain reaction (see 1942).

In December, **Marjorie Courtenay-Latimer**



Echolocation in bats

Bats hunt for food by listening for echoes of their high-pitched calls bouncing off their prey. The larger the prey, the greater is the maximum detection distance.

January Carlo Perrier and Emilio Segrè make the first artificial element, **technetium**

March Hans Krebs shows that **citric acid** keeps tissues alive

British scientists **Dorothy Crowfoot** (later, Hodgkin) and **John Bernal** describe the structure of sterols using **X-ray crystallography**

September Grote Reber completes construction of the first **radio telescope**

British biologist **William Astbury** produces the first **X-ray diffraction** pattern of DNA, showing it has a regular structure

February 16 US Patent Office registers **polyamide 6-6** for Wallace Carothers at DuPont

May 6 German **Hindenburg** airship bursts into flames, ending rigid airship development

January 8 Russian physicist **Pyotr Kapitsa** describes discovery of **superfluidity**

February 24 A **nylon** bristle toothbrush is the first commercial product to be made with nylon yarn

Early 1938 Donald Griffin and Robert Galambos begin the first experiments on bats that would demonstrate **echolocation**